Full Course Assessment
Graded Assignment • 1h

Due Sep 17, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

1. **Why is Apache Spark a popular choice for enterprises?** 1 / 1 point

○ It comes with built-in cluster management
○ It only supports batch processing
● It provides high-level abstraction for simplifying distributed data processing
○ It handles every data engineering function independently

✓ Nice work

Correct! Spark abstracts complexities, making distributed computing accessible.

2. **Which of the following is NOT a limitation of Apache Spark according to the text?** 1 / 1 point

○ Lack of native cluster management
● Inability to integrate with machine learning libraries
○ Limited integration with proprietary data sources
○ Absence of ACID properties

✓ Nice work

Correct! Spark excels at integrating with machine learning components.

3. **Which method is crucial for handling large datasets efficiently in a data engineering project?** 1 / 1 point

○ Manual data entry
○ Real-time processing only
○ Synchronous database replication
● Batch processing

✓ Nice work

Correct! Batch processing is well-suited for large datasets due to its ability to process data in bulk.

4. **Which architectural pattern does Databricks primarily support?** 1 / 1 point

○ Star Schema
● Medallion Architecture
○ Document Store Architecture
○ Data Warehouse Architecture

✓ Nice work

Correct! Databricks supports Medallion Architecture for organizing data.

5. **How does Databricks Marketplace contribute to the platform's functionality?** 1 / 1 point

○ By managing network security
● By offering third-party data solutions
○ By providing real-time database management
○ By hosting code repositories

✓ Nice work

Correct! It brings in external solutions for diverse needs.

6. **What functionality does Databricks provide for task management?** 1 / 1 point

● Scheduling Tasks
○ Data Export
○ Cluster Management
○ SQL Optimization

✓ Nice work

Correct! Workflows manage task sequences and schedules.

7. **What must be provided during Microsoft Azure account creation?** 1 / 1 point

● Credit or debit card details
○ Type of cloud services required
○ A professional email address
○ Company registration number

✓ Nice work

Correct! Payment details are needed for verification despite the free account option.

8. **What is the primary use of the left-side menu in the Databricks UI?** 1 / 1 point

○ To manage billing settings
● To navigate workspace functions
○ To set language preferences
○ To access Azure portal settings

✓ Nice work

Correct! The menu is critical for navigating workspace functionalities.

9. **Which method allows external access to the Databricks control plane?** 1 / 1 point

○ SQL Queries
○ Command Line Interface (CLI)
○ VNC
● REST APIs

✓ Nice work

Correct! REST APIs provide access to Databricks control plane functionalities.

10. **What determines the default payment currency in Azure portal settings?** 1 / 1 point

○ Account type selected
○ Selected default region
○ Preferred language setting
● Location setting

✓ Nice work

Correct! The user's geographical location sets the default currency.

11. **What pricing tier would you choose in Azure Databricks to access all features?** 1 / 1 point

● Premium
○ Enterprise
○ Basic
○ Standard

✓ Nice work

Correct! The Premium tier has all features available.

12. **Which cluster configuration is suitable for cost-effective learning in Databricks?** 1 / 1 point

○ Multi-node cluster
● Single node cluster
○ GPU-enabled cluster
○ High-memory cluster

✓ Nice work

Correct! Single node clusters offer a cost-effective solution for learning purposes.

13. **What is the significance of preinstalled libraries in the Databricks runtime?** 1 / 1 point

○ They depend on specific user permissions to activate
● They enable immediate use of common functionalities
○ They improve the runtime performance significantly
○ They restrict the usage to predefined functions only

✓ Nice work

Correct! They allow developers to utilize standard libraries without setup.

14. **What does the 'FS' utility in Databricks allow you to do?** 1 / 1 point

○ Database management actions
○ Image processing tasks
● File system operations
○ Network monitoring and analysis

✓ Nice work

Correct! The 'FS' utility handles tasks like copy, list, and delete files.

15. **What is the function of auto-scaling in a Databricks cluster?** 1 / 1 point

○ Automatically upgrades cluster hardware
○ Schedules regular maintenance downtime
● Adjusts the number of worker nodes based on demand
○ Changes cluster configurations to meet security standards

✓ Nice work

Correct! Auto-scaling increases or decreases worker nodes as needed.

16. **Why are parameterized notebooks a powerful feature in Databricks?** 1 / 1 point

○ They allow the execution of Python scripts only
● They enable code reusability with different inputs
○ They automatically optimize resource allocation
○ They enhance security by encoding all variables

✓ Nice work

Correct! Parameterization allows the same code to be used with varied data inputs.

17. **What is the primary purpose of the Databricks File System (DBFS)?** 1 / 1 point

● To provide a file interface for scalable data storage
○ To manage user access permissions
○ To set up a database schema
○ To configure Spark clusters

✓ Nice work

Correct! DBFS acts as a scalable storage layer accessible via a file interface.

18. **What is a potential risk of improperly configuring a mount in DBFS?** 1 / 1 point

○ Data processing becomes slower
○ It increases costs exponentially
● Unauthorized users may access sensitive data
○ It consumes more storage than expected

✓ Nice work

Correct! An improper configuration can expose data to unauthorized access.

19. **In Databricks, how can you secure client ID and secret values needed for mounting storage?** 1 / 1 point

○ Save in a local database
○ Encrypt using base64
● Use Azure Key Vault
○ Store in plaintext in code

✓ Nice work

Correct! Azure Key Vault securely manages sensitive information.

20. **When performing the first step to create a storage account in Azure for mounting in DBFS, where do you initiate the process from?** 1 / 1 point

● From Azure portal home page
○ From a local script file
○ From the command line interface
○ Directly in the Databricks UI

✓ Nice work

Correct! The process begins in the Azure portal home page.